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Design and prototyping of an active knee orthosis: A dynamic and mechanical study

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Abstract This study presents the design and prototyping of an active knee orthosis to assist rehabilitation process through the controlled flexion-extension of the joint. 3D scanning and additive manufacturing to create a personalized and lightweight device was used. A dynamic analysis to evaluate the forces and torques acting on the orthosis during movement was conducted. Also, finite element simulations to ensure the mechanical integrity of critical components under certain loads. The manufacturing process included the use of 3D-printed components and a mechanical braking system to adjust the range of motion. Preliminary testing confirmed the orthosis's ability to support the flexion and extension of the device from 0 to 140 degrees. These findings demonstrate the potential of digitally manufactured orthoses to provide customized and effective solutions for lower limb rehabilitation.

1. Introduction

Motor disabilities in the lower extremities seriously affect mobility, impacting on the quality of life of millions worldwide. In Mexico, for example, about 7.1 million people report some disability, and 41 % of them have difficulty walking or climbing stairs [1]. That is almost half their everyday problems, such as moving around at home or outside.

Orthoses are commonly used to assist in the rehabilitation of individuals with neuromusculoskeletal deficiencies [2-4]. Knee orthoses, in particular, help restore and stabilize joint movement during rehabilitation processes for conditions such as ligament injuries [5], muscle weakness [6, 7], or recovery after surgery [8, 9]. However, economic and social factors limit access to this type of device.

Although advances in biomedical engineering have greatly improved orthosis technology in recent decades, high cost [10], complexity [11], and lack of customization remain significant barriers to access [12]. Active or motorized orthoses have been developed [13, 14], but their expensive designs prevent them from being available to all patients, especially in regions with fewer economic resources [15]. Furthermore, this is a severe problem because, curiously, those who could benefit most from these technologies have the least access to them. This is a paradox that this study attempts to solve.

To address these challenges, this research seeks to design and prototype an active knee orthosis that is accessible, affordable, and customizable using digital manufacturing technologies such as 3D scanning and 3D printing. Additive manufacturing has already demonstrated its potential to offer customized solutions in the manufacturing of medical devices [16]. However, many of the current solutions are limited by their mechanical complexity or because they cannot be precisely adjusted to each patient, affecting their effectiveness in rehabilitation [17].

This study introduces an active knee orthosis with a customizable stiffness adjustment mechanism. It is designed to assist with flexion-extension movements, allowing patients to adjust the stiffness and angular displacement based on the rehabilitation needs prescribed by a physiotherapist [18]. Additionally, using 3D scanning for precise fitting and 3D printing for rapid prototyping, which is quite efficient, offers a scalable and personalized solution for knee rehab-

ilitation.

The aim is to address the lack of customizable orthotic devices by developing an effective solution that meets the functional demands of knee rehabilitation. The study presents the design process, including material selection and the structural analysis performed through finite element (FE) simulations to assess mechanical performance.

2. Orthosis analysis

2.1 Knee orthosis

The knee orthosis developed in this project was designed to assist rehabilitation by controlling knee flexion and extension in an angular range between 0° and 140° . A quality function development (QFD) matrix was used to determine the critical design priorities, emphasizing range of motion, stiffness adjustment, and device weight.

Initially, several design ideas were considered, such as a four-bar mechanism and crank-slider system, but ultimately, the choice landed on a simple revolute mechanism with a mechanical braking system.

The proposed design consists of three subassemblies that work together to control knee motion and ensure adjustability for rehabilitation exercises. Fig. 1 shows the overall subassemblies that comprise the entire active knee orthosis. These subassemblies include the primary mechanism, an auxiliary articulation, and a subassembly for fastening molds. The integration of these subassemblies provides the necessary stability and a personalized fit through 3D-printed components.

The mechanical braking system, as illustrated in Fig. 2, plays a special role in adjusting the rigidity of the orthosis. This mechanism allows for precise modulation of stiffness during knee flexion and extension exercises, enabling the orthosis to lock at specific angles based on the patient's rehabilitation needs. The braking system not only enhances the device's functionality but also simplifies the overall design, ensuring ease of use for both the patient and the physiotherapist.

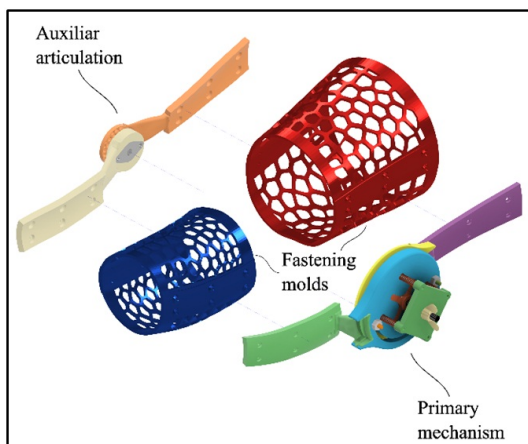


Fig. 1. Design of the active knee orthosis.

2.2 Statics analysis

A static analysis was conducted using the principle of rotational equilibrium to determine the minimum torque needed to move the knee orthosis. The aim was to determine the applied torque τ_{apl} required to overcome both friction and gravity by acting on the orthosis and the patient's leg.

The free body diagram (FBD) that includes the loading forces, boundary conditions and angular displacement variables is illustrated in Fig. 3.

For this analysis, the yellow rigid body is totally constrained, that is, its displacements and rotations in all directions are zero. The external forces include the weight of the orthosis $\vec{W}_o$ and the weight of the patient's leg $\vec{W}_p$. Additionally, two torques are considered: the first is the externally applied torque τ_{apl} resulting from the force exerted by the leg, and the second is the frictional torque τ_f from the braking mechanism, which acts as a reaction to the applied torque and is modeled as in studies [19, 20]. Therefore, the equation for static moment equilibrium (1) is expressed as:

$$\sum \vec{M}_A = \vec{\tau}_f + \vec{\tau}_{apl} + \vec{r}_o \times \vec{W}_o + \vec{r}_p \times \vec{W}_p = 0. \quad (1)$$

Here:

- $\vec{\tau}_f$ is the torque due to friction,
- $\vec{\tau}_{apl}$ is the applied torque by the user,
- $\vec{r}_o \times \vec{W}_o$ is the torque generated by the orthosis weight,
- $\vec{r}_p \times \vec{W}_p$ is the torque generated by the leg's weight.

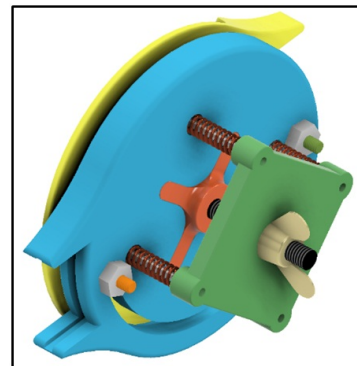


Fig. 2. Design of the mechanical braking system.

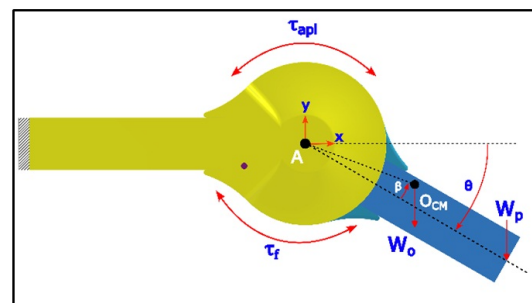


Fig. 3. Free body diagram of the system.

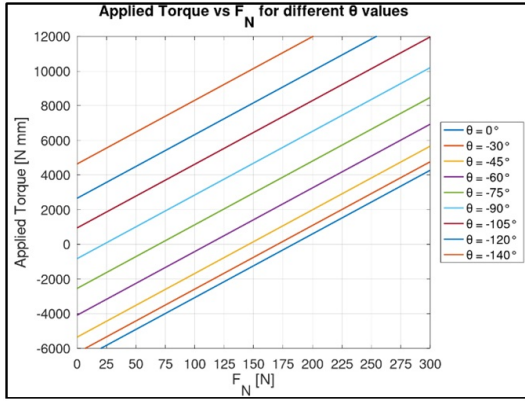


Fig. 4. Applied torque vs normal force for different angles of flexion.

Solving vectorially, the magnitude of the applied torque τ_{apl} , was determined by the Eq. (2) as:

$$\tau_{apl} = \frac{1}{2} \mu_k F_N (R + r) - W_o L_o \cos(\theta + \beta) - \dots - W_p L_p \cos(\theta + \beta). \quad (2)$$

Here:

- μ_k is the kinetic friction coefficient between the contact surfaces,
- F_N is the normal force applied by the springs,
- R and r are the outer and inner radius of the brake disk,
- W_o and W_p are the weights of the orthosis and the leg,
- L_o and L_p are the distances from the rotation axis to the centers of mass of the orthosis and leg,
- θ and β are the angles considered in the analysis.

The outcome of this analysis is shown in Fig. 4, which illustrates the linear relationship between the applied torque τ_{apl} and the normal force F_N at various flexion angles θ . The graph shows the direct relation between the normal force and the torque at several flexion angles; indicating that higher normal forces are required at higher flexion angles to prevent the joint from moving. This means that the torque required to disrupt equilibrium increases as the flexion angle increases.

This analysis provides useful insights for tweaking orthoses according to the user's needs. Also, this allows the physiotherapist to adjust the normal force to control the angular movement of the joint.

2.3 Dynamics analysis

The dynamic analysis of the knee orthosis focuses on the behavior of the system under specific loading and movement conditions. To understand the interaction between forces and torques during motion, the moment equation considering the inertial effects, followed by the calculation of applied torques during motion was developed. This section presents the fundamental equations that model the system dynamics and discusses the results obtained from the analysis.

2.3.1 Kinetic moment equation

The equation of moments was reformulated to include the inertial effects of the system, which are crucial for accurately modeling the dynamic behavior of the orthosis, as is expressed in the Eq. (3):

$$\sum \vec{M}_A = \vec{\tau}_f + \vec{\tau}_{apl} + \vec{r}_o \times \vec{W}_o + \dots \dots + \vec{r}_p \times \vec{W}_p = I_{zz} \ddot{\alpha} + m \ddot{a} L. \quad (3)$$

Where:

- I_{zz} is the mass moment of inertia of the rotating parts relative to point A,
- $\ddot{\alpha}$ is the angular acceleration vector $[0, 0, \ddot{\theta}(t)]^T$,
- m is the total system mass,
- $\ddot{a}$ is the linear acceleration of the system's center of mass,
- L is the distance from point A to the center of mass of the system.

Eq. (3) was reformulated to incorporate both rotational inertia I_{zz} and translational effects, resulting in Eq. (4):

$$\tau_{apl} = \frac{1}{2} \mu_k F_N (R + r) - \dots - (W_o L_o + W_p L_p) \cos(\theta + \beta) - (I_{zz} + mL^2) \ddot{\theta}(t). \quad (4)$$

Where:

- $\ddot{\theta}(t)$ is the angular acceleration profile.

This formulation provides a complete description of the balance between moments, including inertial contributions from both rotation and translation.

2.3.2 Torque calculation during movement

The angular acceleration equation derived from a fifth-degree polynomial profile guaranteeing a smooth transition in motion, was used for the solution of the applied torque. The angular acceleration was modeled in Eq. (5) as:

$$\ddot{\theta}(t) = -1.05t + 0.15744t^2 - 0.0052t^3. \quad (5)$$

Finally, considering all the terms, the applied torque equation yields Eq. (6):

$$\tau_{apl} = \frac{1}{2} \mu_k F_N (R + r) - \dots - (W_o L_o + W_p L_p) \cos(-0.175t^3 + 0.01312t^4 - 0.00026t^5 + \beta) - \dots - (I_{zz} + mL^2) (-1.05t + 0.15744t^2 - 0.0052t^3). \quad (6)$$

This equation allows for the dynamic torque to be evaluated as a function of time, normal force, and the angular acceleration profile. The fifth-degree polynomial was selected for its ability to produce smooth, continuous motion that avoids abrupt changes in acceleration, which could negatively impact on the

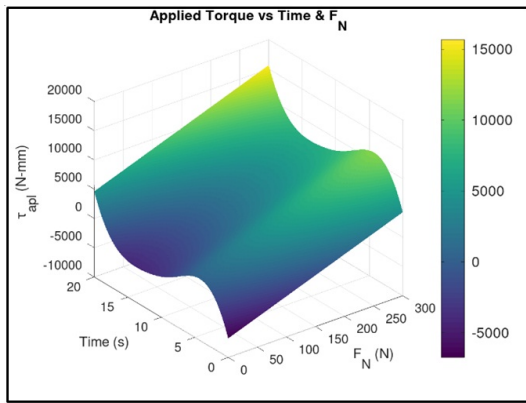


Fig. 5. Torque response surface.

performance and safety of the orthosis during rehabilitation exercises.

2.3.3 Torque response surface

The resulting torque values were plotted to visualize the relationship between the applied torque, time, and normal force. Fig. 5 shows the response surface of the torque over time and under varying normal force conditions. The maximum torque reached approximately 15000 N-mm, while the design limit considered in finite element analysis (FEA) was 12000 N-mm, indicating that under certain extreme conditions, structural limits may be exceeded.

The surface exhibits almost a linear trend between the normal force and time, while a non-linear behavior between the applied torque and time is observed. This non-linearity is attributed to the inertial effects and angular acceleration profile used, as seen in Eq. (6). Peaks in torque reflect moments when both normal force and inertial contributions are significant, highlighting critical points for design consideration.

3. Orthosis modeling and prototyping

3.1 Materials

The active knee orthosis was primarily manufactured using acrylonitrile butadiene styrene (ABS) and stainless-steel components. ABS was chosen for its easy access, good mechanical properties and adaptability to additive manufacturing.

The mechanical properties were determined from the stress-strain curve using a uniaxial tensile test performed according to ASTM D638 standard. Fig. 6 shows the stress-strain curve from where Young modulus is 1.24 GPa, yield strength is 19.21 MPa and tensile strength is 21.04 MPa. Also, it can be observed that elasto-plastic with a clear transition from the elastic to plastic zone is exhibited by the material.

For the orthosis's critical load-bearing parts, such as the shafts and bearings, stainless steel was used due to its high tensile strength (500–700 MPa) and resistance to wear. These components ensure the long-term structural integrity of the

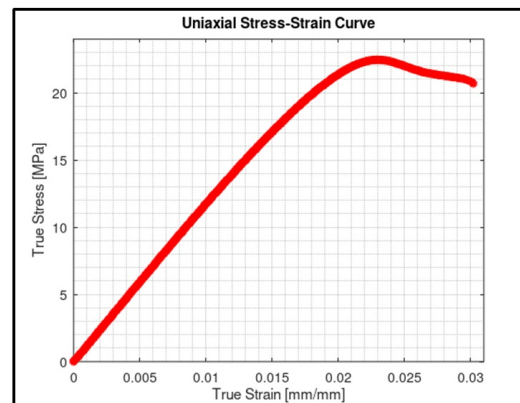


Fig. 6. Uniaxial stress-strain curve for the ABS material.

orthosis, especially under repetitive dynamic stresses.

3.2 Mechanical evaluation by FE

To assess the structural integrity under various loading conditions, the mechanical evaluation of the knee orthosis components was conducted using FEA. This section details the modeling, boundary conditions, and results for critical components, ensuring that the orthosis can withstand expected operational forces without failure.

3.2.1 Model creation and material properties

All components were evaluated using linear static analysis through FEA. However, two critical components were of particular interest in the study: the braking ring and the tibial disc. The mechanical properties were set based on the stress-strain curve presented in Fig. 6. A free mesh with tetrahedral elements of 1 mm were defined based on a convergence study that accurate results. These are shown in Figs. 7 and 8. For both parts, the loading and boundary conditions represent the typical operational condition during use, as follows:

Brake ring: A normal load of 100 N distributed in the points where the compression springs acts were applied. The bottom surface was totally fixed with translational and rotational constraints. This simulated the friction force necessary to immobilize the joint under load.

Tibial disc: The disc experienced the maximum torque of 12000 N-mm at full flexion (140°). Thus, tangential forces of 175 N and 240 N representing the reaction forces between the tibial bar and knee, during flexion and extension process were used.

3.2.2 Numerical results

The numerical analysis provided insights into the displacements, stresses, and safety factors for each component. In the case of the brake ring, the maximum displacement observed was 0.00359 mm, with a safety factor of 8.08. The effective stresses were well below the material yield strength, with a

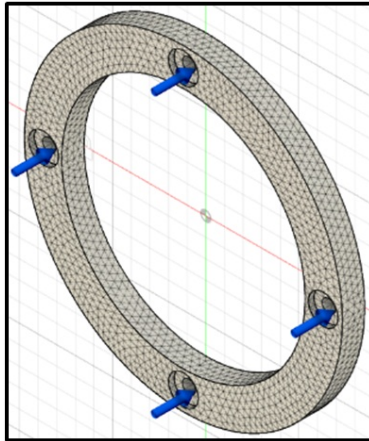


Fig. 7. Mesh and load conditions for braking ring.

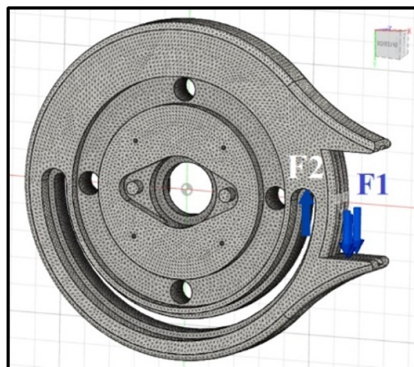


Fig. 8. Mesh and load conditions for tibial disc.

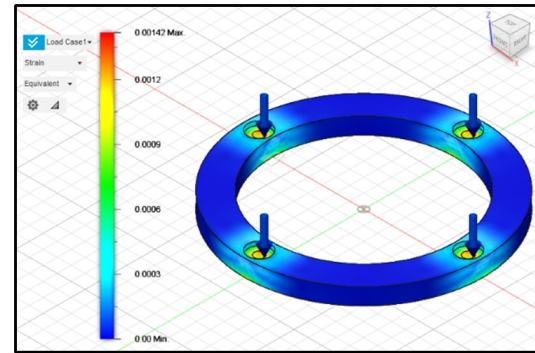
maximum value of 2.47 MPa. The results confirmed that the brake ring would remain structurally safe under the applied loads (see Fig. 9).

For the tibial disc, a maximum strain of 0.0125 and a safety factor of 0.996 were calculated, indicating potential stress concentrations (see Fig. 10(c)). These concentrations were mitigated through design iterations that modified the disc geometry. The maximum effective stress reached 20.074 MPa, higher than for the brake ring but still within the tolerable range for localized stress points.

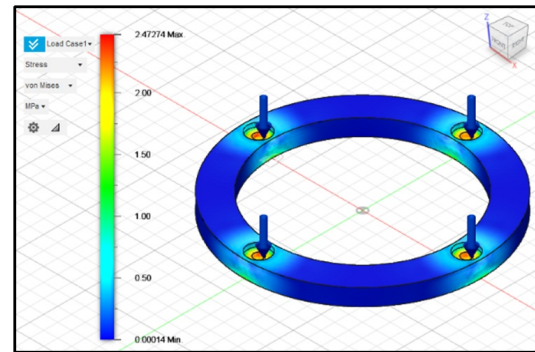
The results from the FEA demonstrate that the key components of the orthosis can withstand the operational loads without compromising structural integrity. Design refinements ensured safety and reliability, making further optimization unnecessary for the analyzed components.

3.3 Manufacturing and assembly

The manufacturing of the knee orthosis was completed using 3D printing technology, with ABS as the primary material due to its mechanical properties and ease of use in additive manufacturing. The orthosis components, including the molds and various connecting parts, were designed to optimize assembly and functionality.



(a) Strain



(b) Effective stress

Fig. 9. Results of FEA for brake ring.

3.3.1 3D printing process

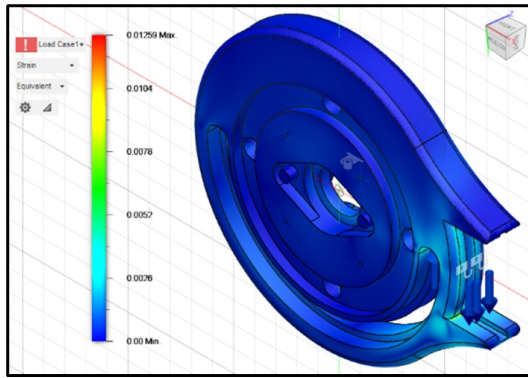
The orthosis prototyping was performed using fused deposition modeling (FDM), with specific attention to printing parameters to ensure high-quality output. The extruder temperature was set to 230 °C, and the print bed was heated to 110 °C. A print speed of 60 mm/s was maintained, while the enclosure of the printer helped stabilize the temperature and minimize warping effects. The design allowed for a dimensional tolerance of +0.3 mm to compensate for material shrinkage after cooling.

Post-processing steps included the removal of support structures and sanding the parts for a smoother finish. Machining was applied where necessary, such as threading the connection points to accommodate screws for the assembly.

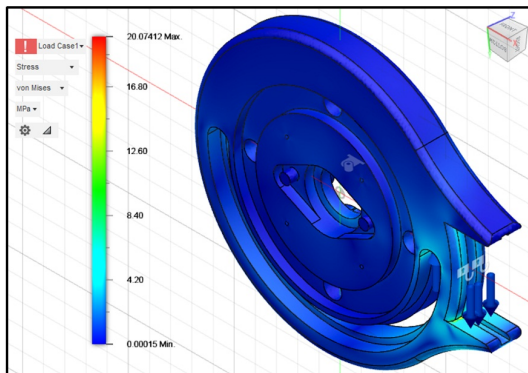
3.3.2 Assembly process

The assembly process followed a multi-stage approach. First, the main mechanism is composed by assembly the steel shaft to the flange bearing and connecting it to the tibial disk. Springs and guide rods were installed in the disk, followed by the external structure, which was adjusted to compress the springs. The auxiliary articulation was then connected using shafts and bearings, ensuring smooth rotation and stability. Finally, the upper and lower molds for the leg were assembled and joined to the primary and auxiliary mechanisms using screws.

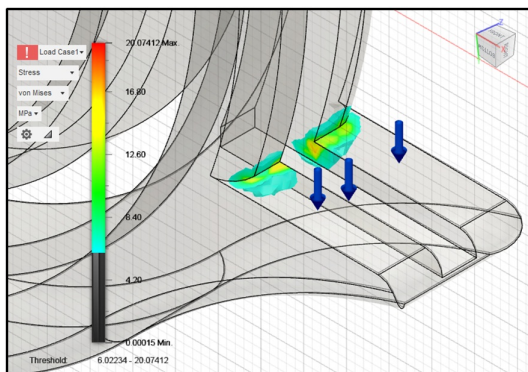
The final verification of the assembly (Fig. 11) ensured that the orthosis was properly fitted and functional, achieving the



(a) Strain



(b) Effective stress



(c) Stress concentrator

Fig. 10. Results of FEA for tibial disc.

desired motion control for knee rehabilitation.

4. Discussion

The results obtained for designing and prototyping a knee orthosis intended for rehabilitation are presented in four relevant areas: dynamic analysis, structural analysis, prototyping, and future work.

First, physical design parameters were established through dynamic analysis so that the braking mechanism would function correctly. This analysis considered the normal force exerted and the angle at which the orthosis was positioned as the

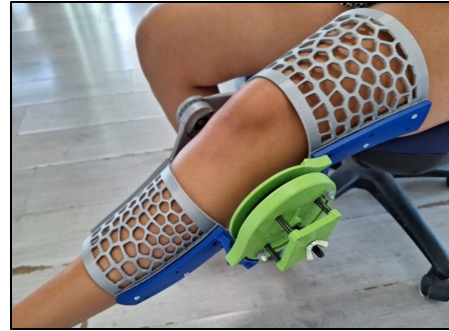


Fig. 11. Final prototype.

input parameters. From this, the torque requirements necessary to be considered in the design of the mechanism and the structural analysis were adequately calculated to adjust the rigidity according to the user's needs. The torque response under dynamic conditions, modeled with a fifth-degree polynomial profile, demonstrated that the device-maintained stability and smooth movement when the transitions were sufficiently slow. Maximum torques reaching 15000 N-mm were found. This peak slightly exceeded the design threshold of 12000 N-mm. However, it provided valuable information to determine what adjustments were needed.

Subsequently, the structural safety of each element and piece that makes up the orthosis was validated by conducting a structural analysis of the device. Each part was simulated using FE analysis under the operational loads defined by the dynamic analysis. As mentioned in the text, the most critical components were the braking ring and tibial disc; however, through the qualitative analysis of the parts, it was determined that the safety factors were adequate according to the functions they were going to perform, except for the stress concentrator explicitly located in the tibial disc, where for future work a possible risk of failure can be successfully reduced.

Prototyping using 3D scanning and additive manufacturing highlights the adaptability to any anthropometric parameter and the precision with which the orthosis prototype can be built. Using ABS allows for relatively fast (compared to previous techniques) and customized production, while post-processing techniques mainly ensure dimensional accuracy. The assembly procedure, created to allow the physiotherapist to make quick and simple adjustment to ensure patient comfort if the mechanism needed to be assembled and disassembled, shows how this design approach can be scaled for customized production, making it a practical and economically viable choice for wider rehabilitation applications.

Finally, future work and opportunities aim to improve structural strength, look for materials that provide better solutions regarding the skin's response, and incorporate control mechanisms to improve the patient's experience. Future work could benefit from incorporating sensors to monitor joint angles and torque in real-time, allowing physiotherapists to directly adapt the numerical values calculated in the dynamic analysis section for rehabilitation.

5. Conclusions

From the presented information the following conclusions are addressed:

A customizable and functional knee orthosis to offer accessible rehabilitation solutions was successfully developed.

The dynamic analysis based on design parameters provides important information that controls structural strength in knee orthosis.

Finite element analyses confirmed the safe operation of each component under defined loads, with the braking ring and tibial disc identified as the most critical elements. Although stress concentration remains in the tibial disc, refining its design can reduce failure risk.

3D scanning and additive manufacturing methods could produce a fully customized orthosis prototype in a flexible, precise, and economical way of creating the device. This highly scalable approach is expected to improve rehabilitation outcomes in different clinical environments.

Continued improvements and exploration of advanced features will further expand its impact and effectiveness in clinical rehabilitation settings.

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Conflict of interest

The authors declare no competing interests.

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